Ching-Hsiang (Adam) Chang

Data Engineer

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github.com/AdamXiang adam-website-tau.vercel.app

Summary

Data engineer with hands-on experience across finance, market research and consumer hardware. I build automated pipelines, medallion data models and dashboards with Python, SQL, Databricks, dbt and Azure. Recent results: a 40-minute ETL job brought under 1 minute per file, 30–80× faster database ingestion, and reporting time cut by 75%. Background in brain science research; currently building and evaluating AI agents with LangGraph.

Experience

Data Engineer, DATARGET — Nov 2025 – Apr 2026

- Led the move from Pandas to parallel processing with ClickHouse Local, bypassing the Python GIL and cutting ETL time from 40 minutes to under 1 minute per file (99% faster).
- Replaced standard SQL inserts with MSSQL BCP for 30–80× faster ingestion; built a Python HandlerFactory (OOP, abstract base classes) for automated schema generation.
- Designed the migration strategy from Azure to Microsoft Fabric and introduced Airflow and dbt with shared code standards.
- Diagnosed geospatial precision errors (latitude/longitude rounding) and time-series logic conflicts, recovering 25× more valid data points for forecasting models.
- Presented a Superset vs. Metabase trade-off analysis to leadership and deployed the chosen tool in Docker for geospatial analytics.

Go-to-Market Intern, Logitech — Jul 2024 – Aug 2024

- Built an automated Tableau dashboard giving stakeholders real-time visibility into product metrics for go-to-market decisions.
- Engineered a smart Google Sheets form with the design team that eliminated manual entry in the creative workflow.

Data Analyst Intern, uppeta — Apr 2024 – Jun 2024

- Consulted on 5+ client projects, turning user behaviour data into strategies that improved user registration.
- Automated internal data processing with Google Apps Script, removing manual bottlenecks.

Data Engineer Intern, CXG — Jul 2023 – Apr 2024

- Engineered a real-time Power BI and Azure Synapse reporting solution with operations managers, reducing reporting time by 75%.
- Architected and standardised a medallion architecture (bronze, silver, gold) as the single source of truth for analysts.
- Orchestrated ELT and HTTP API ingestion pipelines in Azure Data Factory, ensuring 100% on-time scheduled refreshes.
- Migrated legacy Excel workflows to Metabase with optimised SQL transformations on Azure Synapse.

Data Analyst Intern, Cathay Financial Holdings — Feb 2023 – Jul 2023

- Designed a FHIR-standard ER schema that reduced inter-hospital data exchange latency by 75%.
- Built an end-to-end medical LINE bot (Flask, MongoDB) with automated scraping pipelines; designed ChatGPT prompt workflows to automate XML-to-JSON conversion of legacy records.

Selected projects github.com/AdamXiang

- **AdmGPT**: self-hosted ChatGPT-style assistant on Claude and LangGraph with streaming chat, RAG over uploaded documents, web search and long-term memory.
- **AI Agent Evaluation**: end-to-end evaluation of a tool-using refund agent with DeepEval across answer, tool-call and trace-level metrics.
- **Banking CDC pipeline**: PostgreSQL → Debezium/Kafka → MinIO → Snowflake with SCD Type 2 marts, orchestrated by Airflow and dbt.
- **Real-time e-commerce processing with Flink**: sub-100 ms event processing with exactly-once semantics and dual sinks to PostgreSQL and Elasticsearch.
- **Weather analytics on Apache Iceberg**: Spark medallion pipeline handling corrections, late events and duplicates with idempotent MERGE INTO.
- **Kitchen Clash** (kitchen-clash.onrender.com): browser pickleball game with online multiplayer and voice chat.

Skills

Languages	Python, SQL, PySpark
Data engineering	Apache Airflow, dbt, Azure Data Factory, Kafka, Spark, Flink, Delta Live Tables, medallion architecture, CDC
Platforms	Databricks, Azure Synapse, Microsoft Fabric, Snowflake, ClickHouse, SQL Server, MongoDB, Docker
Analytics	Power BI, Tableau, Metabase, Apache Superset, Google Sheets, Excel
AI engineering	LangGraph, LangChain, RAG, Model Context Protocol, DeepEval

Education

M.S., Brain Science — 2021 – 2024
National Yang Ming Chiao Tung University

B.S., Animal Science — 2017 – 2021
National Taiwan University

Certification and service

Databricks Certified Data Engineer Associate

Diplomatic Alternative Service, Taiwan Technical Mission in Eswatini, Nov 2024 – Jul 2025. Supported a national pig industry improvement project.